

## Physics-Informed Denoising Diffusion Probabilistic Models

Technische Universität Berlin

PIML seminar, 2025

## Outline

## I Denoising Diffusion Probabilistic Models (DDPMs)

## Motivation

## Background Concepts

## Structure of DDPMs

## Mathematical Foundations of DDPMs

## Applications

## II Physics-Informed DDPM

## Motivation

## Noise prediction

## Consideration of PDE and data

## Simplification of training objective

## Experimental Results

## Reference

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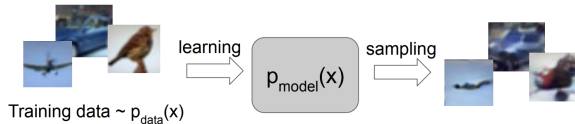
## Simplification of training objective

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# Motivations: Generative Models (GM)

- **Goal:** Given training data, generate new samples from same distribution.

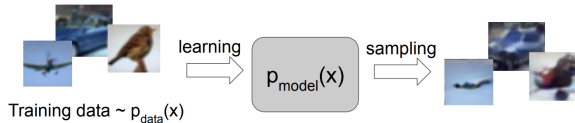


Generative Models [6]

- Objectives:
  - **Density estimation:** Learn  $p_{\text{model}}(\mathbf{x})$  that approximates  $p_{\text{data}}(\mathbf{x})$
  - **Sampling:** How can we generate novel data from the model distribution, i.e.,  $\mathbf{x}_{\text{new}} \sim p_{\text{model}}(\mathbf{x})$  ?

# Motivations: Generative Models (GM)

- **Goal:** Given training data, generate new samples from the same distribution.



Generative Models [6]

- **How can we learn the distribution  $p_{\text{model}}$ ?**
  - **Approach 1:** Maximize the expected log-likelihood of  $p_{\text{model}}(x)$ :  $\max \log p(x)$ .
  - **Approach 2:** Since direct maximization can be too complex, we can instead maximize the **Evidence Lower Bound (ELBO)** of  $\log p(x)$ .

**Fréchet Inception Distance (FID)** is a metric used to assess the quality of images generated by models. **Lower is better.**

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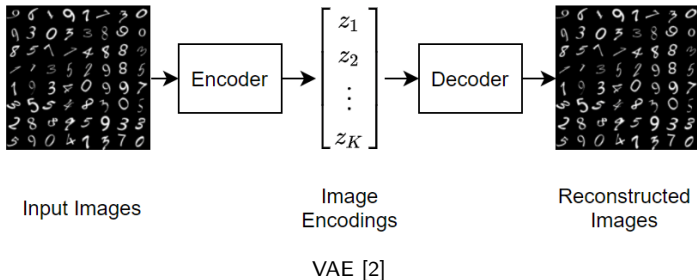
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# Variational Autoencoder (VAE)

- **Encoder:** Maps input data to a distribution in the latent space.
- **Decoder:** Maps samples from the latent space back to the data space.



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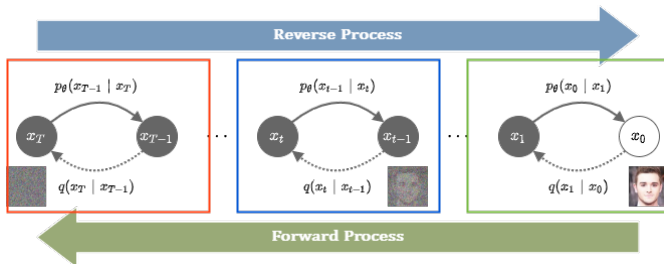
## Simplification of training objective

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## Structure of DDPMs

- ▶ Can be viewed as a special case of VAE: Multiple latent variables with dimensions matching the data.
  - ▶ **Initial Block**  $x_0$  to  $x_1$ :  $x_0$  would be the original data
  - ▶ **Transition Block**  $x_t$  to  $x_{t+1}$ : Markovian transitions.
  - ▶ **Final Block**  $x_{T-1}$  to  $x_T$ : latent  $x_T$  follows a standard Gaussian distribution.
- ▶ **Encoder**  $\rightarrow q(x_t | x_{t-1})$  (**Forward Process**) Gradually adds noise to the data.
- ▶ **Decoder**  $\leftarrow p_\theta(x_{t-1} | x_t)$  (**Backward Process**) Gradually de-noises the data.



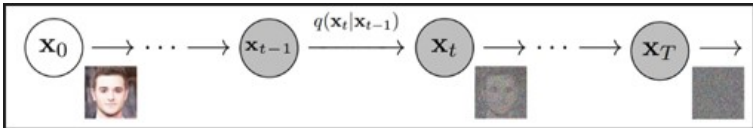
Diffusion Model [2]

## Forward Process (FP)

- **Objective:** progressively corrupt the data until it approximates the pure Gaussian noise at the final timestep  $T$ .
  - The forward process is defined as

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t) \mathbf{I}).$$

- How does  $\alpha_t$  do? How does  $\alpha_t$  influence the transformation? What happens when  $\alpha_t$  is close to 0?



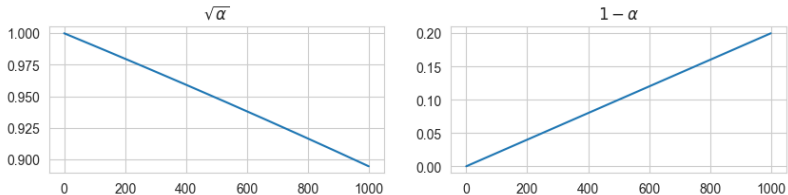
**Forward Process**  $\rightarrow$  (data to noise) [5]

## Forward Process (FP)

- **Objective:** progressively corrupt the data until it approximates the pure Gaussian noise at the final timestep  $T$ .
  - The forward process is defined as

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t) \mathbf{I}).$$

- $(1 - \alpha_t)$  controls the noise variance,  $\sqrt{\alpha_t}$  scales the previous data  $\mathbf{x}_{t-1}$ , ensuring the signal contribution is gradually reduced without shrinking excessively.



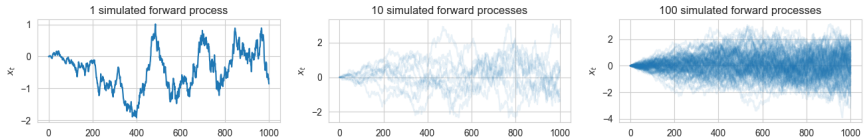
Evolution of the parameters

# Forward Process (FP)

**Recall:** The **forward process** is defined as:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t) \mathbf{I}).$$

While unique forward processes yield random variables at each step  $t$ , the density  $q(\mathbf{x}_t | \mathbf{x}_{t-1})$  at  $t$  is always a parameterized Gaussian:



Simulation of Forward Processes

## Forward Process (FP)

**Recall:** The **forward process** is defined as:

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t) \mathbf{I}).$$

where  $1 - \alpha_t$  is a **variance schedule** that determines the *rate of noise addition*.

**By re-parameterization:**  $\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_{t-1}$ ,

**Questions:**

- Is it possible to express  $\mathbf{x}_t$  in terms of the initial state  $\mathbf{x}_0$ ? How?

# Transition Distribution vs. Conditional Distribution

## Transition Distribution

Let the data sample  $\mathbf{x}_0 \sim q(\mathbf{x})$  represent the original input (e.g., an image). We initialize the **forward process** with  $\mathbf{x}_0$ . The transition from  $\mathbf{x}_{t-1}$  to  $\mathbf{x}_t$  is defined as:

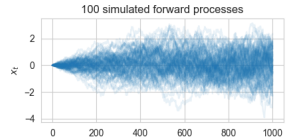
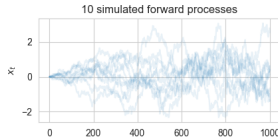
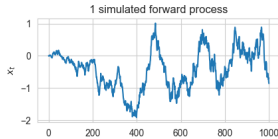
$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t | \sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t) \mathbf{I}).$$

## Conditional Distribution

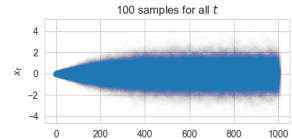
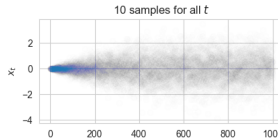
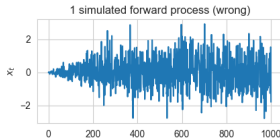
Instead of sampling step-by-step for each  $t$ , we can directly write the **forward process** for distribution any step  $t$  as:

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t | \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad \bar{\alpha}_t = \prod_{s=1}^t \alpha_s.$$

# Transition Distribution vs. Conditional Distribution



Forward Processes with Transition Distribution



Forward Processes with Conditional Distribution

Both approaches lead to the same Gaussian distribution for large enough samples that leads to the observed equivalence of their underlying distributions.

Why Use  $\mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t}\mathbf{x}_{t-1}, (1 - \alpha_t)\mathbf{I})$ ?

**Background:** In **FP**, the final distribution is assumed to be  $\mathcal{N}(\mathbf{x}_T; 0, \mathbf{I})$ .

Suppose  $q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t | a\mathbf{x}_{t-1}, b^2\mathbf{I})$ , then:

$$\mathbf{x}_t = a\mathbf{x}_{t-1} + b\boldsymbol{\epsilon}_{t-1}, \quad \boldsymbol{\epsilon}_{t-1} \sim \mathcal{N}(0, \mathbf{I}).$$

**By recursion:**

$$\mathbf{x}_t = a^t \mathbf{x}_0 + b \underbrace{\left[ \boldsymbol{\epsilon}_{t-1} + a\boldsymbol{\epsilon}_{t-2} + a^2\boldsymbol{\epsilon}_{t-3} + \cdots + a^{t-1}\boldsymbol{\epsilon}_0 \right]}_{\stackrel{\text{def}}{=} \mathbf{w}_t}.$$

**Covariance at  $t \rightarrow \infty$ :**

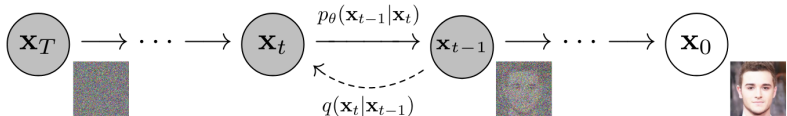
$$\lim_{t \rightarrow \infty} \text{Cov}[\mathbf{w}_t] = \frac{b^2}{1 - a^2} \mathbf{I}, \quad \text{and} \quad \frac{b^2}{1 - a^2} = 1.$$

**Parameterization:** If  $a = \sqrt{\alpha}$ , then  $b = \sqrt{1 - \alpha}$ , leading to:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t}\mathbf{x}_{t-1}, (1 - \alpha_t)\mathbf{I}).$$

## Backward Process (BP)

- **Objective:** Reconstruct the original data by removing noise step by step.
  - Intuitively, gradually **undo** the **forward process** and the goal of diffusion model is to **learn** the reverse denoising process



**Backward Process**  $\rightarrow$  (noise to data) [5]

## Backward Process (BP)

- The **backward or reverse process** is expressed by

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1} | \mu_q(\mathbf{x}_t, \mathbf{x}_0), \sigma_q^2(t)\mathbf{I})$$

which is a Gaussian distribution and fully characterized by its mean and covariance, as follows:

$$\begin{aligned} \mu_q(\mathbf{x}_t, \mathbf{x}_0) &= \frac{(1 - \bar{\alpha}_{t-1})\sqrt{\alpha_t}}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{(1 - \alpha_t)\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_t} \mathbf{x}_0, \\ \Sigma_q(t) &= \frac{(1 - \alpha_t)(1 - \sqrt{\bar{\alpha}_{t-1}})}{1 - \bar{\alpha}_t} \mathbf{I} = \sigma_q^2(t)\mathbf{I}, \end{aligned}$$

where  $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$  accumulates the noise schedule.

## Backward Process (BP)

**Recall:** The **backward or reverse process** is expressed by

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1} | \mu_q(\mathbf{x}_t, \mathbf{x}_0), \sigma_q^2(t)\mathbf{I})$$

- **Problem:**  $\mu_q(\mathbf{x}_t, \mathbf{x}_0)$  requires knowledge of  $\mathbf{x}_0$  (unavailable during RP)
- **Solution:** approximate  $\mathbf{x}_0$  using a neural network  $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)$  parametrized by  $\theta$ :

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \mu_\theta(\mathbf{x}_t, t), \sigma_q^2(t)\mathbf{I}),$$

where

$$\begin{aligned} \mu_\theta(\mathbf{x}_t, t) &= \frac{(1 - \bar{\alpha}_{t-1})\sqrt{\alpha_t}}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{(1 - \alpha_t)\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_t} \hat{\mathbf{x}}_\theta(\mathbf{x}_t), \\ \Sigma_q(t) &= \frac{(1 - \alpha_t)(1 - \sqrt{\bar{\alpha}_{t-1}})}{1 - \bar{\alpha}_t} \mathbf{I} = \sigma_q^2(t)\mathbf{I}, \quad \bar{\alpha}_t = \prod_{s=1}^t \alpha_s. \end{aligned}$$

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# Evidence Lower Bound (ELBO) for DDPMs

**Objective Function:** The loss function is the Evidence Lower Bound (ELBO).

- Its ELBO is defined as:

$$\begin{aligned}
 \text{ELBO}_{\theta}(\mathbf{x}_0) &= - \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} [D_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t))] \\
 &= - \sum_{t=1}^T \frac{1}{2\sigma_q^2(t)} \frac{(1 - \alpha_t)^2 \bar{\alpha}_{t-1}}{(1 - \bar{\alpha}_t)^2} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left[ \left\| \hat{\mathbf{x}}_0(\mathbf{x}_t) - \mathbf{x}_0 \right\|^2 \right] \\
 &= - \sum_{t=1}^T \frac{1}{2\sigma_q^2(t)} \frac{(1 - \alpha_t)^2 \bar{\alpha}_{t-1}}{(1 - \bar{\alpha}_t)^2} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left[ \left\| \hat{\mathbf{x}}_0 \left( \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon_t \right) - \mathbf{x}_0 \right\|^2 \right].
 \end{aligned}$$

# Evidence Lower Bound (ELBO) for DDPMs

- Maximizing the **ELBO**:

$$\begin{aligned}
 & \arg \max_{\theta} \sum_{\mathbf{x}_0 \in \mathcal{X}} \text{ELBO}_{\theta}(\mathbf{x}_0) \\
 &= \arg \min_{\theta} \sum_{\mathbf{x}_0 \in \mathcal{X}} \sum_{t=1}^T \frac{1}{2\sigma_q^2(t)} \frac{(1 - \alpha_t)^2 \bar{\alpha}_{t-1}}{(1 - \bar{\alpha}_t)^2} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left\| \hat{\mathbf{x}}_0 \left( \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon_t \right) - \mathbf{x}_0 \right\|^2 \\
 &= \arg \min_{\theta} \sum_{\mathbf{x}_0 \in \mathcal{X}} \sum_{t=1}^T \frac{1}{M} \sum_{m=1}^M \frac{1}{2\sigma_q^2(t)} \frac{(1 - \alpha_t)^2 \bar{\alpha}_{t-1}}{(1 - \bar{\alpha}_t)^2} \left\| \hat{\mathbf{x}}_0 \left( \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon_t^{(m)} \right) - \mathbf{x}_0 \right\|^2,
 \end{aligned}$$

- $\varepsilon_t^{(m)} \sim \mathcal{N}(0, \mathbf{I})$ , and the summation over  $\mathbf{x}_0 \in \mathcal{X}$  means we consider all samples in the training set  $\mathcal{X}$ .

# Evidence Lower Bound (ELBO) for DDPMs

For simplification, the loss function can be expressed as

$$\mathcal{L}_{\text{simple}}(\theta) = \mathbb{E}_{t, x_0} \left[ \lambda_t \|\hat{x}_\theta(x_t) - x_0\|^2 \right].$$

- ▶ Optimizing go down to learn a neural network  $\hat{x}_0$  to predict the original ground truth image from an arbitrarily noisified version of it [5].
- ▶ The training of this model involves training a **denoiser**  $\hat{x}_0$ .
- ▶ Therefore, the resulting model is known as the **denoising diffusion probabilistic model (DDPM)**.

## Training the Denoiser $\hat{\mathbf{x}}_{\theta}(\cdot)$ [1]

For each image  $\mathbf{x}_0$  in the training dataset, perform the following steps until convergence:

1. Sample a random time step  $t \sim \text{Uniform}[1, T]$ .
2. Generate a noised sample  $\mathbf{x}_t^{(m)}$  by drawing from the distribution:

$$\mathbf{x}_t^{(m)} \sim \mathcal{N}(\mathbf{x}_t \mid \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}),$$

or equivalently:

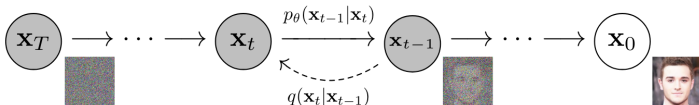
$$\mathbf{x}_t^{(m)} = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon_t^{(m)}, \quad \varepsilon_t^{(m)} \sim \mathcal{N}(0, \mathbf{I}).$$

3. Perform a gradient descent step on the objective:

$$\nabla_{\theta} \left\{ \frac{1}{M} \sum_{m=1}^M \left\| \hat{\mathbf{x}}_{\theta}(\mathbf{x}_t^{(m)}) - \mathbf{x}_0 \right\|^2 \right\}.$$

- $$\mathbf{x}_{t-1} \sim p_{\theta}(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} \mid \mu_{\theta}(\mathbf{x}_t), \sigma_q^2(t)\mathbf{I})$$

- $$\begin{aligned} \mathbf{x}_{t-1} &= \boldsymbol{\mu}_\theta(\mathbf{x}_t) + \sigma_q(t)\boldsymbol{\epsilon}, \quad \text{where } \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}) \\ &= \frac{(1 - \bar{\alpha}_{t-1})\sqrt{\alpha_t}}{1 - \bar{\alpha}_t}\mathbf{x}_t + \frac{(1 - \alpha_t)\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_t}\hat{\mathbf{x}}_\theta(\mathbf{x}_t) + \sigma_q(t)\boldsymbol{\epsilon}. \end{aligned}$$



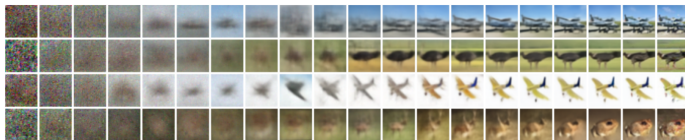
## Reverse Diffusion [5]

# Inference Algorithm

1. Start with random noise  $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$ .
2. Iteratively denoise by calculating  $\hat{\mathbf{x}}_\theta(\mathbf{x}_t)$  using our trained denoiser.
3. Update according to

$$\mathbf{x}_{t-1} = \frac{(1 - \bar{\alpha}_{t-1})\sqrt{\alpha_t}}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{(1 - \alpha_t)\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_t} \hat{\mathbf{x}}_0(\mathbf{x}_t) + \sigma_q(t)\varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \mathbf{I})$$

4. Repeat for  $t = T, T-1, \dots, 1$  to reconstruct the data  $\mathbf{x}_0$ .



Sampling in DDPM (noise to data) [3]

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# DALLE-4 (Text To Image)



Teddy bears mixing sparkling chemicals as mad scientists

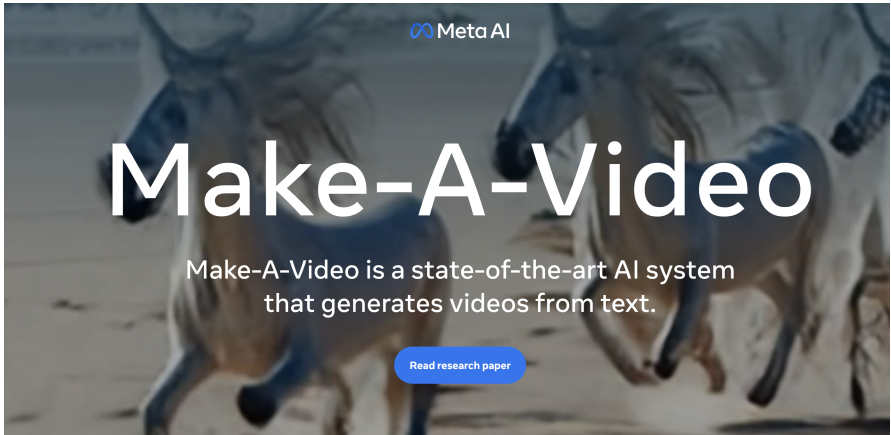


An astronaut riding a horse in a photorealistic style



A bowl of soup as a planet in the universe

Dalle-4



make a video (facebook)

## Conclusion

- **Diffusion Models** significantly enhance image generation quality.
- **Key Processes:** Forward Process & Backward Process
- **Workflow:** Training & Sampling
- **Limitation:** High computational cost due to training and sampling

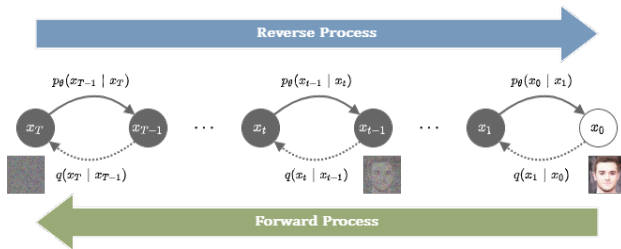


Illustration of the Diffusion Model Process [2]

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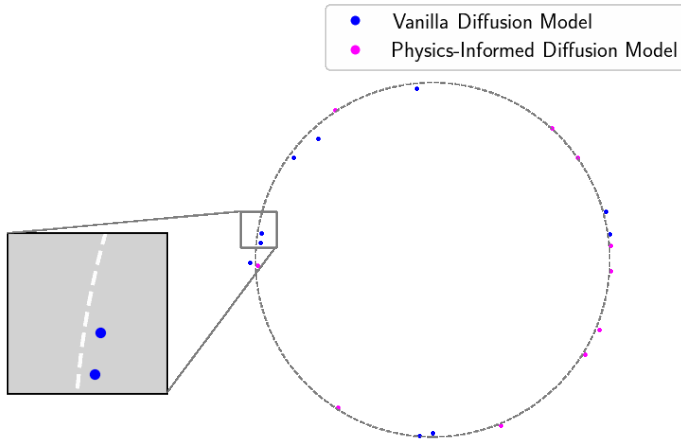
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# Comparison to Physics-Informed Diffusion Models

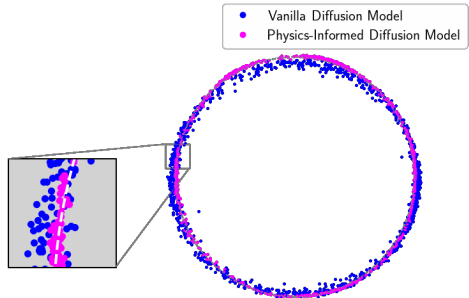


# Motivation

**Goal:** Incorporate physical laws into Denoising Diffusion Probabilistic Models (DDPMs) to generate solutions consistent with governing equations, such as PDEs.

## Motivation

- ▶ Ensure generated outputs respect physical principles (e.g., conservation laws).
- ▶ Improve robustness and generalization for scientific and engineering tasks.
- ▶ Solve inverse problems with uncertainty quantification.



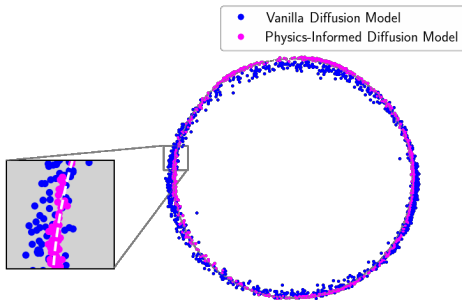
Results differences between Physics-Informed and Vanilla approach

### Ideas:

- ▶ Data only indirectly ensure results will comply with physical laws
- ▶ use PDEs as an *infinite* source of information

### Methods:

- ▶ Keep data based training of the DDPM
- ▶ For inference implement initial boundary condition and PDE in the total loss function.



### Results differences between Physics-Informed and Vanilla approach

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# Analogy of predicting noise and clear image

## Recall:

- Optimization goal: minimize Kullback-Leibler divergence

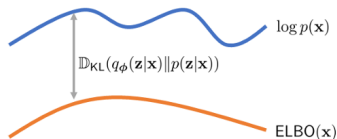
$$\mathbb{D}_{KL} = \frac{1}{2\sigma_q^2(t)} \|\mu_q(x_0, x_t) - \mu_\theta(x_t)\|^2$$

- the reverse mean distribution is a function of  $\mathbf{x}_t$  and  $\mathbf{x}_0$

$$\mu_q(x_0, x_t) = \frac{(1 - \bar{\alpha}_{t-1})\sqrt{\alpha_t}}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{(1 - \alpha_t)\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_t} \mathbf{x}_0$$

$$\mu_\theta(\mathbf{x}_t) = \frac{(1 - \bar{\alpha}_{t-1})\sqrt{\alpha_t}}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{(1 - \alpha_t)\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_t} \hat{\mathbf{x}}_\theta(\mathbf{x}_t)$$

- model will predict  $\hat{\mathbf{x}}_\theta$  close to the clear image  $\mathbf{x}_0$ .



## Residue-type Approaches

We apply the **residue-type approach** to predict noise  $\varepsilon$  instead of the image  $\mathbf{x}_0$ . This means learning  $\varepsilon$  directly.

### Residuals in Physics-Informed Models:

- ▶ Residuals  $\mathcal{R}(\mathbf{x}_0)$  measure deviations from the exact PDE solution.
- ▶ Treating  $\mathcal{R}(\mathbf{x}_0)$  as a random variable enforces physics constraints probabilistically.

### Noise in Diffusion Models:

- ▶ In DDPMs, noise  $\varepsilon_t$  follows a Gaussian distribution.
- ▶ The denoiser  $\varepsilon_\theta$  learns to correct  $\varepsilon_t$  during the reverse process.

### Connection:

- ▶ Residuals in physics-informed models and noise in diffusion models both represent deviations from a target.
- ▶ Minimizing residuals in physics-informed models parallels denoising in DDPMs:

$$\min \mathbb{E}[\|\mathcal{R}(\mathbf{x}_0)\|^2] \quad \sim \quad \min \mathbb{E}[\|\varepsilon_t - \varepsilon_\theta\|^2].$$

## Analogy of predicting noise and clear image

- Use the relation between  $\mathbf{x}_0$  and  $\varepsilon_0$ :

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon_0,$$

to reformulate  $\mu_q(\mathbf{x}_t, \mathbf{x}_0)$

$$\mathbf{x}_0 = \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \varepsilon_0}{\sqrt{\bar{\alpha}_t}}$$

$$\mu_q(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t} \sqrt{\alpha_t}} \varepsilon_0.$$

- In words, we converted  $\mu_q(\mathbf{x}_t, \mathbf{x}_0)$  from a function of  $\mathbf{x}_0$  to a function of  $\varepsilon_0$ .
- Similarly, predicted mean estimator  $\mu_\theta(\cdot)$  is defined by

$$\mu_\theta(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t} \sqrt{\alpha_t}} \hat{\varepsilon}_\theta(\mathbf{x}_t).$$

## A New Evidence Lower Bound (ELBO)

**Objective Function:** The new loss function for noise prediction is defined by:

- ▶ The new ELBO is given by:

$$\begin{aligned}\text{ELBO}_{\theta}(\mathbf{x}_0, \varepsilon_0) &= - \sum_{t=1}^T \frac{1}{2\sigma_q^2(t)} \frac{(1 - \alpha_t)^2 \bar{\alpha}_{t-1}}{(1 - \bar{\alpha}_t)^2} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left\| \hat{\varepsilon}_{\theta}(\mathbf{x}_t) - \varepsilon_0 \right\|^2 \\ &= - \sum_{t=1}^T \frac{1}{2\sigma_q^2(t)} \frac{(1 - \alpha_t)^2 \bar{\alpha}_{t-1}}{(1 - \bar{\alpha}_t)^2} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left\| \hat{\varepsilon}_{\theta}(\sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \alpha_t} \varepsilon_0) - \varepsilon_0 \right\|^2.\end{aligned}$$

- ▶ The new ELBO depends on  $\mathbf{x}_0$  and  $\varepsilon_0$ .
- ▶ A simplified loss function is:

$$\mathcal{L}_{\text{simple}}(\theta) = \mathbb{E}_{\mathbf{x}_0, \varepsilon} \left[ \lambda_t \left\| \hat{\varepsilon}_{\theta}(\mathbf{x}_t) - \varepsilon_0 \right\|^2 \right].$$

- ▶ To train the denoiser, we solve the optimization problem:

$$\arg \min_{\theta} \mathbb{E}_{\mathbf{x}_0, \varepsilon_0} \text{ELBO}_{\theta}(\mathbf{x}_0, \varepsilon_0) \approx \arg \min_{\theta} \sum_{\mathbf{x}_0 \sim \mathcal{X}} \frac{1}{M} \sum_{m=1}^M \text{ELBO}_{\theta}(\mathbf{x}_0, \varepsilon_0^{(m)}),$$

- ▶ Using SGD, we iteratively update  $\theta$  as:

$$\theta \leftarrow \theta - \eta \nabla_{\theta} \left\| \hat{\varepsilon}_{\theta}(\mathbf{x}_t) - \varepsilon_0 \right\|^2,$$

where  $\eta$  is the learning rate.

# Outline

## I Denoising Diffusion Probabilistic Models (DDPMs)

Motivation

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Motivation

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**Consideration of PDE and data**

Simplification of training objective

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## Reference

Physical laws are generally formulated as PDEs over a domain  $\Omega \subset \mathbb{R}^c$

$$\mathbf{F}[\mathbf{u}(\boldsymbol{\xi})] = 0, \quad \boldsymbol{\xi} = (\xi_1, \xi_2, \dots, \xi_d)^T, \quad \mathbf{u} = (u_1(\boldsymbol{\xi}), u_2(\boldsymbol{\xi}), \dots, u_c(\boldsymbol{\xi}))^T \in \mathbb{R}^c$$

$$\mathbf{B}[\mathbf{u}(\boldsymbol{\xi})] = 0, \quad \boldsymbol{\xi} \in \partial\Omega$$

$\mathbf{F}$  differential operator,  $\mathbf{B}$ , Boundary condition,  $\mathbf{u}$  sought solution field,  $\xi$  dimensions with dependency

- Introduction of residual  $R(\mathbf{x}_0)$  of a solution field  $\mathbf{x}_0$  by concatenating  $\mathbf{F}[\mathbf{x}_0]$  and  $\mathbf{B}[\mathbf{x}_0]$ , for example  $R_{MAE}(\mathbf{x}_0)$

# Consideration of PDE

Steps to consider PDEs in optimization objective of log likelihood:

- ▶ Introduce residual  $\hat{r}$  as virtual observable
- ▶ sample from distribution:  $q_{\mathcal{R}}(\hat{r}|x_0) = \mathcal{N}(\hat{r}; \mathcal{R}(x_0), \sigma^2 I)$
- ▶ compute virtual likelihood

$$p_{\theta}(\hat{r}) = \int p_{\theta}(\hat{r}, x_0) dx_0 = \int q_{\mathcal{R}}(\hat{r}|x_0) p_{\theta}(x_0) dx_0 = \mathbb{E}_{x_0 \sim p_{\theta}(x_0)} q_{\mathcal{R}}(\hat{r}|x_0)$$

- ▶ substitute  $p_{\theta}(\hat{r})$  in the usual log-likelihood maximizer

$$\arg \max_{\theta} \mathbb{E}_{\hat{r}}[\log p_{\theta}(\hat{r})] = \arg \max_{\theta} \mathbb{E}_{x_0 \sim p_{\theta}(x_0)} [\log q_{\mathcal{R}}(\hat{r} = 0|x_0)]$$

## Consideration of PDE and data

- ▶ Introduce residual  $\hat{r}$  as virtual observable
- ▶ sample from distribution:  $q_{\mathcal{R}}(\hat{r}|x_0) = \mathcal{N}(\hat{r}; \mathcal{R}(x_0), \sigma^2 I)$
- ▶ compute virtual likelihood

$$p_{\theta}(\hat{r}) = \int p_{\theta}(\hat{r}, x_0) dx_0 = \int q_{\mathcal{R}}(\hat{r}|x_0) p_{\theta}(x_0) dx_0 = \mathbb{E}_{x_0 \sim p_{\theta}(x_0)} q_{\mathcal{R}}(\hat{r}|x_0)$$

- ▶ substitute  $p_{\theta}(\hat{r})$  in the usual log-likelihood maximizer

$$\arg \max_{\theta} \mathbb{E}_{\hat{r}}[\log p_{\theta}(\hat{r})] = \arg \max_{\theta} \mathbb{E}_{x_0 \sim p_{\theta}(x_0)}[\log q_{\mathcal{R}}(\hat{r} = 0|x_0)]$$

- ▶ identifying solutions that satisfy governing equations (PDE) is in general non trivial
  - ▶ available data of  $x_0$  as guidance towards feasible solutions
- ▶ extend optimization objective by data-based training objective (noise prediction)

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## II Physics-Informed DDPM

- Motivation
- Noise prediction
- Consideration of PDE and data
- Simplification of training objective**
- Experimental Results

## Reference

## Simplification of training objective

Joint loss is costly due to many iterations while sampling

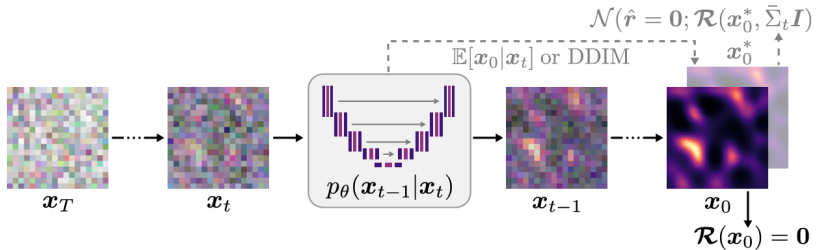
- ▶ **Mean estimation:** minimize a (weighted) mean-squared distance between predicted and true sample
  - ▶ to mitigate inconsistencies during sample steps it is proposed to increase variance of introduced residual likelihood with increasing  $t$
- ▶ **sample estimation:** evaluate residual of actual sample  $x_0$
- ▶ introduce  $x_0^*$  as either mean or sample estimate, the PDE residual optimization objective then reads:

$$\mathbb{E}_{x_{1:T} \sim p_\theta(x_{1:T})} [\log q_{\mathcal{R}}(\hat{r} = 0 | x_0^*(x_{1:T}))]$$

- ▶ model  $q_{\mathcal{R}}(\hat{r} | x_0^*(x_{1:T}))$  as variance scheduler of the reverse process

$$q_{\mathcal{R}}(\hat{r} | x_0^*(x_{1:T})) = \prod_{t=1}^T q_{\mathcal{R}}(\hat{r} | x_0^*(x_t, t)) \text{ where } q_{\mathcal{R}}(\hat{r} | x_0^*(x_t, t)) = \mathcal{N}(\hat{r}; \mathcal{R}(x_0^*), \Sigma_t / cI)$$

# Simplification of training objective



Physics-informed denoising process illustrated [4]

## Simplification of training objective

- recall simple loss function for PDDMs

$$L_{\text{simple}}(\theta) = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} \left[ \lambda \|\mathbf{x}_0 - \mathbf{x}_0^*(\mathbf{x}_t(\mathbf{x}_0, \epsilon), t)\|^2 \right]$$

- changed to noise prediction loss function

$$L_{\text{simple\_noise}}(\theta) = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} \left[ \lambda_t \|\hat{\epsilon}_\theta(\mathbf{x}_t) - \epsilon_0\|^2 \right]$$

- now extend to physics informed loss

$$L_{\text{PIDM}}(\theta) = \mathbb{E}_{t \sim [1, T], \mathbf{x}_0: T \sim q(\mathbf{x}_0: T)} \left[ \lambda_t \|\hat{\epsilon}_\theta(\mathbf{x}_t, t) - \epsilon_0\|^2 + \frac{1}{2\bar{\Sigma}_t} \|\mathcal{R}(\mathbf{x}_0^*(\mathbf{x}_t, t))\|^2 \right]$$

- with  $\bar{\Sigma}_t = \Sigma_t / c$

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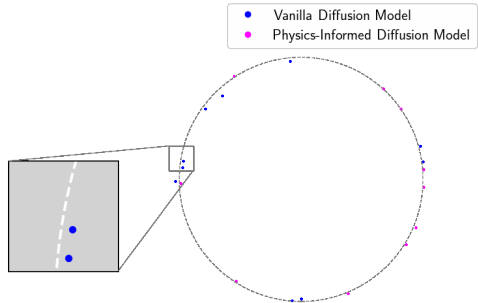
Experimental Results

## Reference

## Experimental Results of toy problem

**Objective:** learn distribution  $q(x)$  for  $S_1 = \{\xi \in \mathbb{R}^2 : \|\xi\| = 1\}$

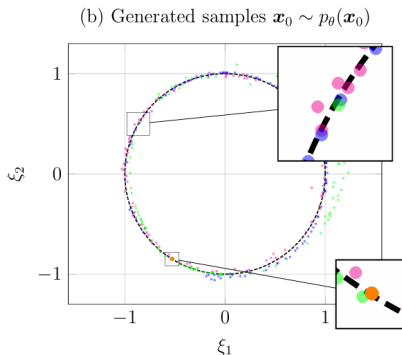
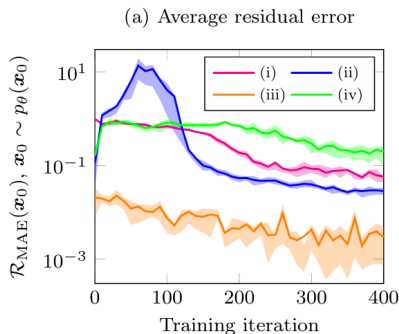
- ▶ Compare results of 4 different loss-functions
  - ▶ (i) vanilla approach - variational bound on data likelihood
  - ▶ (ii) proposed PIDM loss
  - ▶ (iii) only considering residual loss
  - ▶ (iv) considering PIDM loss but with uninformative data ( $x_0 \sim \mathcal{N}(0, I)$ )



## Experimental Results of toy problem

**Objective:** learn distribution  $q(x)$  for  $S_1 = \{\xi \in \mathbb{R}^2 : \|\xi\| = 1\}$

- ▶ (i) vanilla
- ▶ (ii) proposed PIDM
- ▶ (iii) only residual loss
- ▶ (iv) PIDM loss with  $x_0 \sim \mathcal{N}(0, I)$



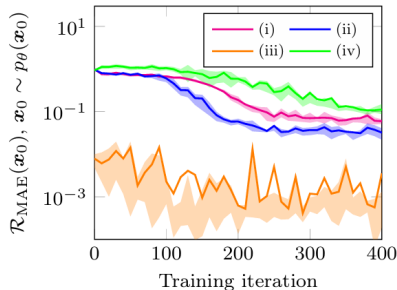
**Mean estimation:** average residual evaluation

## Experimental Results of toy problem

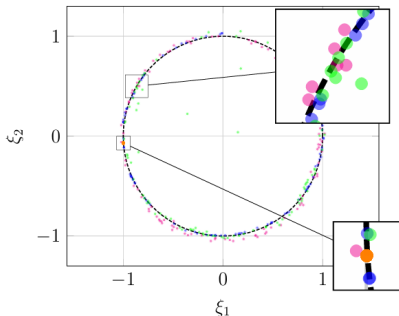
**Objective:** learn distribution  $q(x)$  for  $S_1 = \{\xi \in \mathbb{R}^2 : \|\xi\| = 1\}$

- ▶ (i) vanilla
- ▶ (ii) proposed PIDM
- ▶ (iii) only residual loss
- ▶ (iv) PIDM loss with  $x_0 \sim \mathcal{N}(0, I)$

(a) Average residual error



(b) Generated samples  $x_0 \sim p_\theta(x_0)$



**Sample estimation:** average residual evaluation

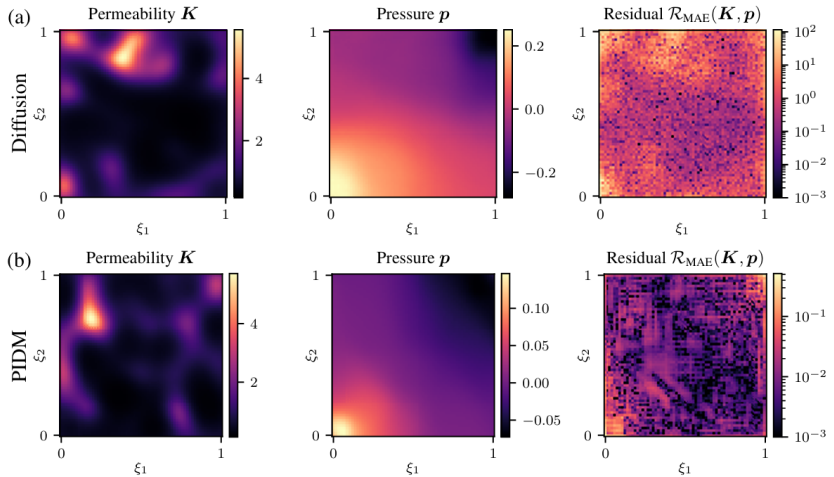
## Example: Darcy flow

- ▶ local 2D flow eq. :

$$\mathbf{q} = -\frac{k}{\mu} \nabla p, \quad \Omega = [0, 1]^2$$

- ▶ sample permeability field  $k$  from Gaussian random field
- ▶ solve governing equations with Finite-Difference-Method (FDM) to create data sets and calculate residuals (training and validation set)
- ▶ create a U-Net architecture with 64x64 grid array for input/output fields
- ▶ choose appropriate parameter  $c$  to increase variance virtual residual likelihood of mean or sample estimation

# Darcy flow results



- TU Berlin, Physics-Informed Denoising Diffusion Probabilistic Models, 2025 ◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ 🔍 ↺ 55/55